

EE 560: Reconfigurable Computing

Course Code	EE 560
Course Name	Reconfigurable Computing
L-T-P-C	3-0-2-4
Prerequisites	EE 210/201P- Digital System Design & Practicum, CS 201/201P Computer Organization & Practicum.
Intended for	PG/UG
Distribution	Elective for M. Tech (VLSI), UG and other PG courses
Approval	39th BoA (Mar 2021); 35th Senate (Jun 2022), in the M.Tech VLSI curriculum

Course Contents

This course has three hour lecture session accompanied by two hours of laboratory session per week. List of course modules are provided below.

- **Introduction to reconfigurable computing:** [3 Lectures]
 - Reconfigurable computing, history and survey, fixed Vs reconfigurable computing, applications of reconfigurable computing
- **FPGA Design:** [6 Lectures]
 - Introduction to FPGA, LUT devices and mapping, partitioning, placement and routing algorithms. A case study on ALU design
- **Reconfigurable computing architectures:** [5 Lectures]
 - Performance evaluation, Coarse grained and fine grained reconfigurable computing, a case study on coarse grained reconfigurable computing, Multi-FPGA architectures, Dynamic reconfiguration, total versus partial reconfiguration
- **Reconfigurable computing:** [2 Lectures]
 - Power Reduction techniques for FPGA
- **Reconfigurable computing applications** [6 Lectures]
 - Distributed Arithmetic, CORDIC algorithm on FPGA
- **Reconfigurable computing-Security to FPGA** [4 Lectures]
 - Physical Unclonable Functions-Introduction and implementation on FPGA, on-chip authentication of FPGA-based devices
- **Hardware-Software Co-design** [4 Lectures]
 - Introduction, Partitioning, Scheduling, Synthesis, Retiming, Pipelining, Unfolding, analysis and Estimation
- **Finite State Machine with Datapath** [3 Lectures]
 - Introduction, finite state machines, finite state machine with datapath (FSMD), a case study on FSMD
- **Hardware-Software Interfaces** [3 Lectures]
 - Hardware software Communication, one way and two way handshake, blocking and non-blocking data transfer, On-chip Bus, Bus transfer and topologies
- **Case study** [6 Lectures]
 - Decoders, Crypto processor, CORDIC processor.

Laboratory Modules

The practicum of this course implements the FPGA based implementation of the algorithms and architectures discussed during the course. Experiments such as UART interfacing with FPGA, CORDIC algorithm, interfacing camera with FPGA, image compression and cryptographic algorithm on FPGA.

Text Books

1. Scott Hauck Andre DeHon, **Reconfigurable Computing: The theory and practice of FPGA based Computation**, Morgan Kaufmann, 2008.
2. F. Vahid and T. Givargis, **Embedded Systems: A Unified Hardware Software Introduction**, John Wiley and Sons, 2011.

References

1. Patrick Schaumont, **A Practical Introduction to Hardware/Software Co design**, Springer, 2010.
2. Bobda, Christophe, **Introduction to Reconfigurable Computing: Architectures, algorithms and applications**, Springer, 2007.
3. I. Koren, **Computer Arithmetic Algorithms**, A.K. Peters Ltd., 2002.
4. Behrooz Parhami, **Computer Arithmetic - Algorithms and Hardware Designs**, 2nd Edition, Oxford university press, 2010.
5. Peter J. Ashenden, **The designer's guide to VHDL**, Morgan Kaufmann, 2008.
6. IEEE research papers on relevant topics.